Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Phillips, William D. Ph.D.		
Project Title:	Evaporative Cooling and Bose Condensates in Microgravity: Picokelvin Atoms In Space		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Gaithersburg	**State:**	MD
Zip Code:	20899-8424	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	02/01/1998	**End Date:**	01/31/2005
No. of Post Docs:	5	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	On Earth, the time available to observe any free system is limited, at least in part, by the time required for the system to fall out of the apparatus. One second is a typical limit for practical experiments. In microgravity, this limitation is greatly increased, and interaction times as long as one thousand seconds may be possible. To develop techniques for achieving BEC or evaporatively cooled atomic gases in microgravity and further cooling such gases to the picokelvin range. The performance of the best atomic clocks is limited by the thermal motion of the atoms. Reduction of this thermal motion leads to better performance, as does the increase in observation time that comes with such reduction. Improved time standards, and particularly their operation in space, can be expected to provide for better deep-space navigation as well as the opportunity for tests of fundamental physical theories.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 06/01/2005)